

EX. 2: Comparative Case Study of Traditional Development and Low-Code Development Using Zoho Creator

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AIM

To study and compare traditional application development with low-code application development using Zoho Creator, focusing on development effort, coding requirements, time, customization, maintenance, and ease of implementation.

OBJECTIVE

1. To understand the basic process of traditional application development.
2. To study the low-code development approach using Zoho Creator.
3. To consider the development of a simple Student Management Application using both approaches.
4. To compare both approaches based on time, coding effort, flexibility, and maintenance.
5. To identify suitable situations for using traditional and low-code development.

SOFTWARE REQUIREMENTS

Requirement	Specification
Traditional Development	HTML, CSS, JavaScript / Python / Java
Low-Code Platform	Zoho Creator
Database	MySQL / PostgreSQL or equivalent
Browser	Google Chrome / Microsoft Edge

Requirement	Specification
Operating System	Windows 10/11
Internet	Required for Zoho Creator

THEORY

Traditional Application Development

Traditional application development requires the developer to write program code for different parts of an application. The user interface, application logic, database connection, validation, and other functions are implemented manually.

The developer normally designs the database, creates forms and web pages, writes backend code, connects the application to the database, and performs testing before deployment. This approach provides greater control over the application and allows developers to implement customized functionality according to specific requirements.

Low-Code Application Development

Low-code development allows applications to be created using visual development tools with minimal manual programming. Components such as forms, reports, and workflows can be configured using drag-and-drop interfaces.

Zoho Creator is a low-code development platform that provides visual tools for creating forms, reports, pages, and workflows. Additional logic can also be implemented using configuration or scripting when required.

CASE STUDY

For comparison, a simple Student Management Application is considered.

The application contains the following modules:

- Student Profile
- Course Enrollment
- Academic Record
- Guardian Contact

The same basic functionality is considered for both traditional development and Zoho Creator.

A. Traditional Development Approach

In the traditional approach, the developer manually implements:

- HTML forms for collecting student information
- CSS for designing the user interface
- JavaScript for client-side validation
- Backend code for processing submitted information
- Database tables for storing student records
- SQL queries for inserting and retrieving information
- Authentication and access control
- CRUD operations for managing student records

This requires separate development and testing of the frontend, backend, and database components.

B. Zoho Creator Low-Code Approach

In Zoho Creator, the application can be developed using the visual form builder.

The general process is:

1. Create a new application in Zoho Creator.
2. Create the required forms using the form builder.
3. Add fields such as Student ID, Name, Department, Email, and Phone Number.
4. Configure the required field properties and validations.
5. Generate reports based on the created forms.
6. Configure workflows for required automated actions.
7. Add scripting only when additional application logic is needed.
8. Test the application using sample records.
9. Deploy the completed application.

PROCEDURE

A. Traditional Development

1. Identify the requirements of the Student Management Application.
2. Design the database structure.
3. Create the required database tables.
4. Develop the frontend using HTML and CSS.
5. Add input validation using JavaScript.
6. Implement the backend functionality.
7. Establish the connection between the application and database.
8. Implement CRUD operations for student records.
9. Test each module.
10. Deploy the completed application.

B. Low-Code Development Using Zoho Creator

1. Login to Zoho Creator.
2. Create a new application.
3. Create the required forms using the visual form builder.
4. Add appropriate fields and configure their properties.
5. Set mandatory fields and validation rules.
6. Create reports to display the stored records.
7. Configure workflows wherever automation is required.
8. Add Deluge scripting if additional logic is needed.
9. Test the application with sample records.
10. Deploy the application.

COMPARATIVE ANALYSIS

Parameter	Traditional Development	Zoho Creator Low-Code
Coding Requirement	More coding required ▾	Minimal coding ▾
Development Time	Generally higher ▾	Generally lower ▾
Technical Skill	Higher ▾	Moderate ▾
UI Development	Manually developed ▾	Visual form builder ▾

Parameter	Traditional Development	Zoho Creator Low-Code
Database Setup	Designed and config... ▾	Platform-supported ▾
Validation	Programmed manually ▾	Configuration and sc... ▾
Workflow	Implemented using c... ▾	Visual workflows ▾
Customization	Very high ▾	High within platform f... ▾
Maintenance	Requires developer e... ▾	Comparatively simpler ▾
Deployment	Requires application ... ▾	Platform-supported ▾
Learning Curve	Higher ▾	Lower ▾
Development Cost	May be higher ▾	Can be lower for sim... ▾
Scalability	Depends on system ... ▾	Depends on platform... ▾
Suitable For	Complex / custom ap... ▾	Rapid business appli... ▾

ADVANTAGES AND DISADVANTAGES

Traditional Development

Advantages:

- Provides complete control over the application design.
- Allows extensive customization.
- Suitable for complex applications.
- Gives flexibility in choosing technologies.
- Allows developers to control the complete application architecture.

Disadvantages:

- Requires stronger programming knowledge.
- Development generally takes more time.
- Testing and debugging require additional effort.

- Database and deployment setup have to be handled separately.
- Maintenance may require continuous developer involvement.

Zoho Creator Low-Code

Advantages:

- Applications can be developed quickly.
- Requires less manual programming.
- Provides ready-made form and UI components.
- Makes form and database creation easier.
- Supports workflow automation.
- Useful for rapid prototyping and business applications.

Disadvantages:

- Customization can be limited compared with fully coded applications.
- Development depends on the features provided by the platform.
- Complex requirements may require additional scripting.
- Platform-specific limitations can affect highly customized applications.

RESULT

Thus, the comparative case study of traditional application development and low-code development using Zoho Creator was studied successfully.

Traditional development provides greater control and customization but generally requires more programming effort and development work. Zoho Creator provides a visual development environment that reduces manual coding and simplifies the creation of forms, reports, and workflows. The comparison helps in understanding how the choice of development approach can depend on the application's requirements, complexity, customization needs, and development effort.